Experiment 2: Online Hotel Room Booking System

Aim

To design and develop an online hotel room booking website using HTML, CSS and JavaScript in which guests can register, check room availability, book, change or cancel bookings (up to 3 days before arrival), view booking history and manage their profile, and the administrator can manage the quantity and price of the four room types, change or cancel any booking, and edit the static information pages.

Algorithm

1. Start and display the home page with the room types and the static pages (Room Info, About Us, Contact Us, Q&A, Travel Guide, Privacy Policy).
2. Registration: validate name, email, 10-digit phone, unique username and password; store the guest.
3. Login: verify username and password for the selected role (Guest / Administrator) and open the matching menu.
4. Guest - Availability: read check-in, check-out and guests; validate check-in >= today and check-out > check-in. For each room type, available = quantity - number of active bookings whose dates overlap (existing check-in < new check-out AND new check-in < existing check-out).
5. Guest - Book: if available > 0, save the booking (status Booked); total = nights x price per night.
6. Guest - Change / Cancel: if (arrival date - today) >= 3 days, allow the change (re-check availability) or set status to Cancelled; otherwise show an error.
7. Guest - View booking history and update the profile (name, email, phone).
8. Admin - Update quantity and price of the Single, Double, Deluxe and Suite rooms.
9. Admin - Edit or cancel any booking without the 3-day restriction.
10. Admin - Select a static page and edit its content; the changes appear on the public page.
11. Logout returns to the home page. Stop.

Program (HTML, CSS and JavaScript) - hotel.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Grand Palm Hotel - Room Booking</title>
<style>
:root{box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);
--bg:#f6f4ef;--card:#fff;--tx:#25221c;--mut:#6f6a5e;--ac:#8a5a14;--acx:#fff;--bd:#ddd6c6;--bad:#b3261e;--ok:#1e7a46}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#17150f;--card:#221f17;--tx:#eee9dc;--mut:#a39d8c;--
    ac:#d9a24a;--acx:#1a1408;--bd:#3b3626;--bad:#f08a80;--ok:#63c48d}}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--tx);font-family:Georgia,'Times New Roman',serif;line-height:1.5}
header{background:var(--ac);color:var(--acx);padding:12px 16px}
header h1{margin:0 0 8px;font-size:1.25rem}
header nav{display:flex;flex-wrap:wrap;gap:4px}
header button{background:transparent;color:inherit;border:1px solid transparent;padding:5px 10px;border-
    radius:5px;cursor:pointer;font:inherit;font-size:.88rem}
header button:hover,header button.on{border-color:currentColor}
main{max-width:980px;margin:16px auto;padding:0 12px}
.card{background:var(--card);border:1px solid var(--bd);border-radius:10px;padding:16px;margin-bottom:14px}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:12px}
button.b{background:var(--ac);color:var(--acx);border:0;border-radius:6px;padding:8px 14px;cursor:pointer;font:inherit;font-
    size:.9rem}
button.b:disabled{opacity:.4;cursor:not-allowed}
button.a{background:transparent;color:var(--tx);border:1px solid var(--bd)}
button.s{padding:3px 9px;font-size:.8rem}
input,select,textarea{padding:8px;border:1px solid var(--bd);border-
    radius:6px;background:var(--bg);color:var(--tx);font:inherit;font-size:.9rem;max-width:100%}
label{font-size:.85rem;color:var(--mut);display:flex;flex-direction:column;gap:3px}
.row{display:flex;gap:10px;flex-wrap:wrap;align-items:flex-end;margin-bottom:10px}
.tw{overflow-x:auto}
table{border-collapse:collapse;width:100%;font-size:.88rem}
th,td{text-align:left;padding:7px 9px;border-bottom:1px solid var(--bd);white-space:nowrap}
th{color:var(--mut)}
h2{margin:0 0 10px;font-size:1.1rem}
.bad{color:var(--bad);font-weight:700}.ok{color:var(--ok);font-weight:700}.mut{color:var(--mut);font-size:.85rem}
.pre{white-space:pre-line}
.price{font-size:1.2rem;color:var(--ac);font-weight:700}
.login{max-width:360px;margin:0 auto}.login input,.login select{width:100%;margin-bottom:8px}
</style>
</head>
<body>
<header><h1>■ Grand Palm Hotel</h1><nav id="nav"></nav></header>
<main id="app"></main>
<script>
const $=id=>document.getElementById(id);
const esc=s=>String(s).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));
const today=()=>new Date().toISOString().slice(0,10);
const add=(d,n)=>{const x=new Date(d);x.setDate(x.getDate()+n);return x.toISOString().slice(0,10)};
const days=(a,b)=>Math.round((new Date(b)-new Date(a))/864e5);
const msg=t=>alert(t);

const D={
 admin:{u:'admin',p:'admin123'},
 users:[{u:'guest',p:'guest123',name:'Anita Rao',email:'anita@mail.com',phone:'9876543210'}],
 rooms:[{t:'Single',qty:5,price:1500},{t:'Double',qty:4,price:2500},{t:'Deluxe',qty:3,price:4000},{t:'Suite',qty:2,price:7000}],
 bk:[{id:1,u:'guest',t:'Double',ci:add(today(),2),co:add(today(),4),g:2,st:'Booked'},
     {id:2,u:'guest',t:'Deluxe',ci:add(today(),20),co:add(today(),23),g:2,st:'Booked'}],
 pages:{
  rooms:['Room Information','Single: one bed, ideal for solo travellers.\nDouble: two beds or one queen bed, for couples or
    friends.\nDeluxe: spacious room with balcony, mini-bar and city view.\nSuite: separate living area, bathtub and premium
    amenities.'],
  about:['About Us','Grand Palm Hotel has welcomed guests since 1998. We offer comfortable rooms, warm service and a central
    location.'],
  contact:['Contact Us','Grand Palm Hotel, 12 Beach Road, Chennai - 600001\nPhone: +91 44 1234 5678\nEmail:
    stay@grandpalm.example'],
  qa:['Customer Service Q&A','Q: What is check-in time?\nA: 12:00 noon. Check-out is 11:00 AM.\n\nQ: Can I change or cancel a
    booking?\nA: Yes, up to 3 days before the arrival date.'],
  travel:['Local Travel & Shipping Guide','Airport: 20 km (30 minutes by taxi).\nRailway station: 5 km.\nNearby: Marina Beach,
    Fort Museum, Kapaleeshwarar Temple.\nWe can arrange luggage shipping on request.'],
  privacy:['Privacy Policy','We collect only the details needed for your booking. Your data is never sold or shared with third
    parties.']
 }
};
let role=null,me=null,view='home',q={},editId=null,nid=10;

const go=v=>{view=v;editId=null;render()};
const nights=b=>days(b.ci,b.co);
const price=t=>D.rooms.find(r=>r.t==t).price;
const
    avail=(t,ci,co,skip)=>D.rooms.find(r=>r.t==t).qty-
    D.bk.filter(b=>b.t==t&&b.st=='Booked'&&b.id!=skip&&b.ci<co&&ci<b.co).length;

function render(){
 const pub=[['home','Home'],['rooms','Room Info'],['about','About Us'],['contact','Contact Us'],['qa','Q&A'],['travel','Travel
    Guide'],['privacy','Privacy']];
 let n=pub;
 if(role=='user')n=n.concat([['book','Check &amp; Book'],['my','My Bookings'],['profile','My Profile']]);
```

```
 else if(role=='admin')n=n.concat([['arooms','Manage Rooms'],['abk','Manage Bookings'],['apages','Edit Pages']]);
 n=n.concat(me?[['out','Logout ('+esc(me.u)+')']]:[['login','Login / Register']]);
 $('nav').innerHTML=n.map(x=>`<button class="${x[0]==view?'on':''}"
    onclick="${x[0]=='out'?'logout()':`go('${x[0]}')`}">${x[1]}</button>`).join('');
 const V={home,login:loginV,book:bookV,my:myV,profile:profV,arooms:aroomsV,abk:abkV,apages:apagesV};
 $('app').innerHTML=(V[view]||(()=>page(view)))();
}
function logout(){role=me=null;go('home')}
function page(k){const p=D.pages[k];return `<div class="card"><h2>${esc(p[0])}</h2><div class="pre">${esc(p[1])}</div></div>`}
function home(){
 return `<div class="card"><h2>Welcome to Grand Palm Hotel</h2><p>Comfortable rooms, central location. Register to book online
    and manage your stay.</p></div>
 <div class="grid">`+D.rooms.map(r=>`<div class="card"><h2>${r.t} Room</h2><div class="price">Rs. ${r.price}<span class="mut"> /
     night</span></div><div class="mut">${r.qty} rooms in total</div></div>`).join('')+'</div>';
}

/* ---- login / register ---- */
function loginV(){
 return `<div class="login"><div class="card"><h2>Login</h2><select id="lr"><option value="user">Guest</option><option
    value="admin">Administrator</option></select>
 <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password"><button class="b"
    onclick="login()">Login</button>
 <p class="mut">Demo: guest / guest123  |  admin / admin123</p></div>
 <div class="card"><h2>Register</h2><input id="rn" placeholder="Full name"><input id="re" placeholder="Email"><input id="rh"
    placeholder="Phone (10 digits)">
 <input id="ru" placeholder="Username"><input id="rp" type="password" placeholder="Password"><button class="b"
    onclick="reg()">Register</button></div></div>`;
}
function login(){
 const r=$('lr').value,u=$('lu').value.trim(),p=$('lp').value;
 const f=r=='admin'?(D.admin.u==u&&D.admin.p==p?D.admin:null):D.users.find(x=>x.u==u&&x.p==p);
 if(!f)return msg('Invalid username or password');
 role=r;me=f;go(r=='admin'?'arooms':'book');
}
function reg(){
 const
    o={name:$('rn').value.trim(),email:$('re').value.trim(),phone:$('rh').value.trim(),u:$('ru').value.trim(),p:$('rp').value};
 if(!o.name||!o.u||!o.p)return msg('Please fill all fields');
 if(!/^\S+@\S+\.\S+$/.test(o.email))return msg('Enter a valid email');
 if(!/^\d{10}$/.test(o.phone))return msg('Phone must be 10 digits');
 if(D.users.some(x=>x.u==o.u))return msg('Username already taken');
 D.users.push(o);msg('Registered successfully. Please login.');
}

/* ---- guest: availability and booking ---- */
function bookV(){
 let h=`<div class="card"><h2>Check Room Availability</h2><div class="row">
 <label>Check-in<input type="date" id="ci" min="${today()}" value="${q.ci||''}"></label>
 <label>Check-out<input type="date" id="co" value="${q.co||''}"></label>
 <label>Guests<input type="number" id="gs" min="1" max="6" value="${q.g||2}" style="width:80px"></label>
 <button class="b" onclick="chk()">Check Availability</button></div></div>`;
 if(q.ci){
  const n=days(q.ci,q.co);
  h+=`<div class="card tw"><h2>Rooms for ${q.ci} to ${q.co} (${n} night${n>1?'s':''})</h2><table><tr><th>Room</th><th>Price /
    night</th><th>Available</th><th>Total</th><th></th></tr>`+
  D.rooms.map(r=>{const a=avail(r.t,q.ci,q.co);return `<tr><td>${r.t}</td><td>Rs. ${r.price}</td><td>${a>0?`<span
    class="ok">${a} left</span>`:'<span class="bad">Sold out</span>'}</td><td>Rs. ${r.price*n}</td>
  <td><button class="b s" ${a?'':'disabled'} onclick="book('${r.t}')">Book</button></td></tr>`}).join('')+'</table></div>';
 }
 return h;
}
function chk(){
 const ci=$('ci').value,co=$('co').value,g=+$('gs').value;
 if(!ci||!co)return msg('Select check-in and check-out dates');
 if(ci<today())return msg('Check-in date cannot be in the past');
 if(co<=ci)return msg('Check-out must be after check-in');
 if(g<1)return msg('Enter number of guests');
 q={ci,co,g};render();
}
function book(t){
 if(avail(t,q.ci,q.co)<1)return msg('Room not available');
 D.bk.push({id:++nid,u:me.u,t,ci:q.ci,co:q.co,g:q.g,st:'Booked'});
 msg('Booking confirmed! Total: Rs. '+price(t)*days(q.ci,q.co));go('my');
}

/* ---- booking list (guest and admin) ---- */
function canChange(b){return role=='admin'||days(today(),b.ci)>=3}
function cancelB(id){
 const b=D.bk.find(x=>x.id==id);
 if(!canChange(b))return msg('Cancellation is allowed only 3 or more days before the arrival date');
 b.st='Cancelled';render();
}
function editB(id){
 const b=D.bk.find(x=>x.id==id);
 if(!canChange(b))return msg('Changes are allowed only 3 or more days before the arrival date');
 editId=id;render();
}
function editForm(){
 const b=D.bk.find(x=>x.id==editId);
 return `<div class="card"><h2>Change Booking #${b.id}</h2><div class="row">
 <label>Room<select id="et">${D.rooms.map(r=>`<option ${r.t==b.t?'selected':''}>${r.t}</option>`).join('')}</select></label>
 <label>Check-in<input type="date" id="eci" value="${b.ci}"></label><label>Check-out<input type="date" id="eco"
    value="${b.co}"></label>
```

```
 <label>Guests<input type="number" id="eg" min="1" max="6" value="${b.g}" style="width:80px"></label>
 <button class="b" onclick="saveEdit()">Save</button><button class="b a" onclick="go(view)">Close</button></div></div>`;
}
function saveEdit(){
 const b=D.bk.find(x=>x.id==editId),t=$('et').value,ci=$('eci').value,co=$('eco').value;
 if(!ci||co<=ci)return msg('Check-out must be after check-in');
 if(role!='admin'&&ci<today())return msg('Check-in date cannot be in the past');
 if(avail(t,ci,co,b.id)<1)return msg('Selected room is not available for those dates');
 Object.assign(b,{t,ci,co,g:+$('eg').value});editId=null;msg('Booking updated');render();
}
function bkTable(list,admin){
 return `<div class="card tw"><h2>${admin?'All Bookings':'My Booking
    History'}</h2><table><tr><th>ID</th>${admin?'<th>Guest</th>':''}<th>Room</th><th>Check-in</th><th>Check-
    out</th><th>Nights</th><th>Total</th><th>Status</th><th></th></tr>`+
 (list.map(b=>`<tr><td>${b.id}</td>${admin?`<td>${esc(b.u)}</td>`:''}<td>${b.t}</td><td>${b.ci}</td><td>${b.co}</td><td>${nights
    (b)}</td><td>Rs. ${price(b.t)*nights(b)}</td>
 <td class="${b.st=='Booked'?'ok':'bad'}">${b.st}</td><td>${b.st=='Booked'?`<button class="b a s"
    onclick="editB(${b.id})">Change</button> <button class="b a s"
    onclick="cancelB(${b.id})">Cancel</button>`:''}</td></tr>`).join('')||'<tr><td colspan="9" class="mut">No
    bookings.</td></tr>')+
 `</table>${admin?'':'<p class="mut">Changes and cancellations are allowed up to 3 days before arrival.</p>'}</div>`;
}
function myV(){return (editId?editForm():'')+bkTable(D.bk.filter(b=>b.u==me.u),false)}
function profV(){
 return `<div class="card"><h2>My Profile</h2><div class="row"><label>Name<input id="pn"
    value="${esc(me.name)}"></label><label>Email<input id="pe" value="${esc(me.email)}"></label>
 <label>Phone<input id="ph" value="${esc(me.phone)}"></label><button class="b" onclick="saveP()">Update</button></div><p
    class="mut">Username: ${esc(me.u)}</p></div>`;
}
function saveP(){
 if(!/^\d{10}$/.test($('ph').value))return msg('Phone must be 10 digits');
 me.name=$('pn').value;me.email=$('pe').value;me.phone=$('ph').value;msg('Profile updated');
}

/* ---- admin ---- */
function aroomsV(){
 return `<div class="card tw"><h2>Room Quantity and Price</h2><table><tr><th>Room type</th><th>Quantity</th><th>Price / night
    (Rs.)</th><th></th></tr>`+
 D.rooms.map((r,i)=>`<tr><td>${r.t}</td><td><input type="number" min="0" id="rq${i}" value="${r.qty}" style="width:80px"></td>
 <td><input type="number" min="1" id="rp${i}" value="${r.price}" style="width:100px"></td><td><button class="b s"
    onclick="saveRoom(${i})">Update</button></td></tr>`).join('')+'</table></div>';
}
function saveRoom(i){
 const q2=+$('rq'+i).value,p=+$('rp'+i).value;
 if(q2<0||p<1)return msg('Enter valid quantity and price');
 D.rooms[i].qty=q2;D.rooms[i].price=p;msg(D.rooms[i].t+' room updated');
}
function abkV(){return (editId?editForm():'')+bkTable(D.bk,true)}
function apagesV(){
 const k=q.pk||'rooms';
 return `<div class="card"><h2>Edit Static Pages</h2><div class="row"><label>Page<select id="pk"
    onchange="q.pk=this.value;render()">`+
 Object.keys(D.pages).map(x=>`<option value="${x}" ${x==k?'selected':''}>${D.pages[x][0]}</option>`).join('')+
 `</select></label></div><label>Content<textarea id="pt" rows="9"
    style="width:100%">${esc(D.pages[k][1])}</textarea></label><br><button class="b" onclick="savePage()">Save
    Page</button></div>`;
}
function savePage(){D.pages[$('pk').value][1]=$('pt').value;msg('Page updated')}

render();
</script>
</body>
</html>
```

Output

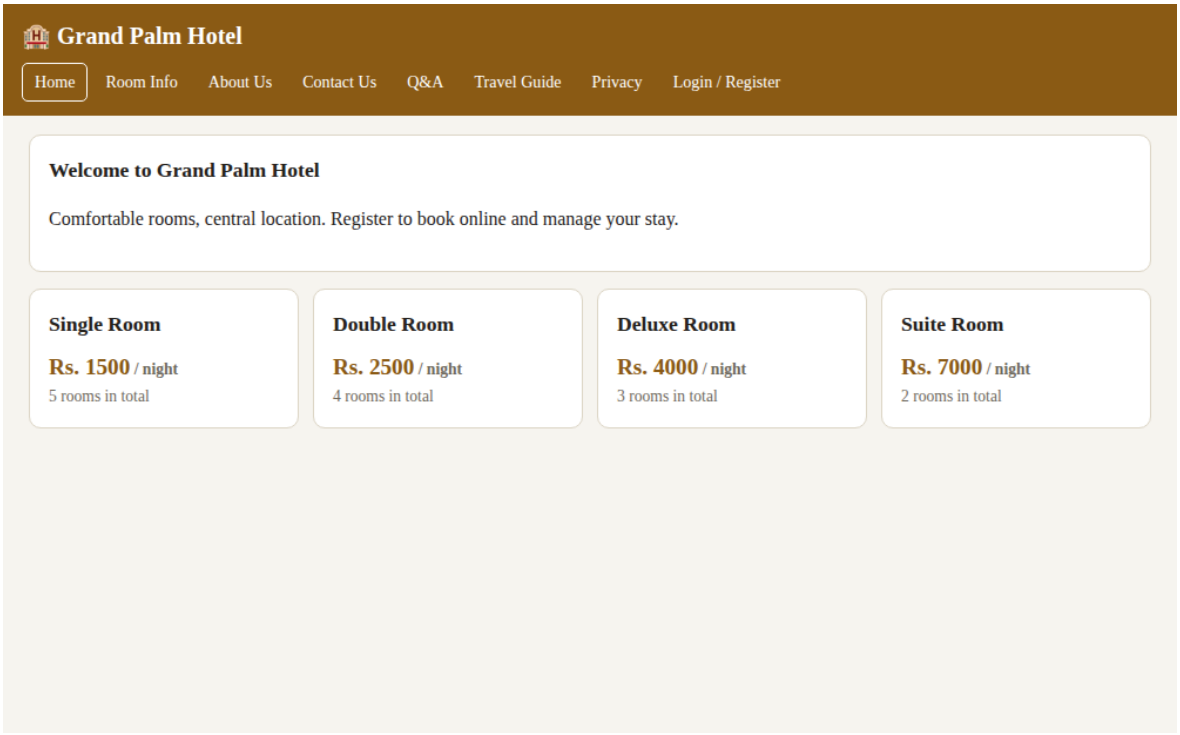


Fig 1: Home page with room types and prices

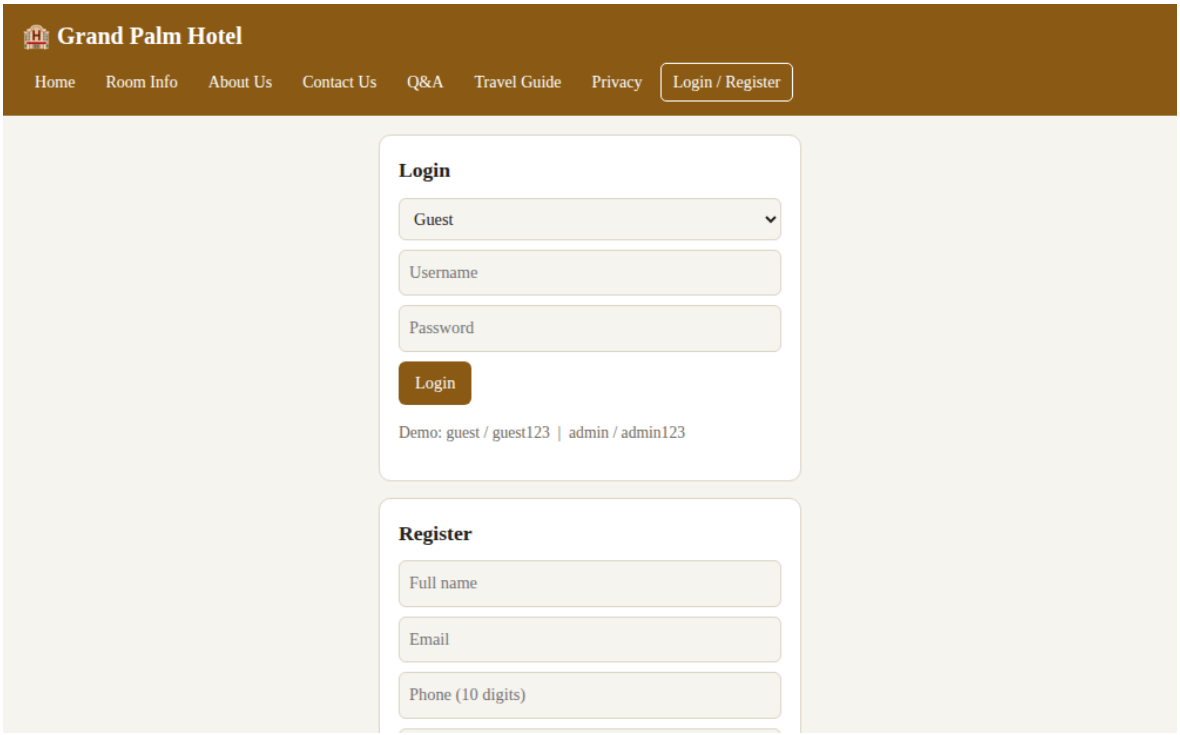


Fig 2: Login and registration page

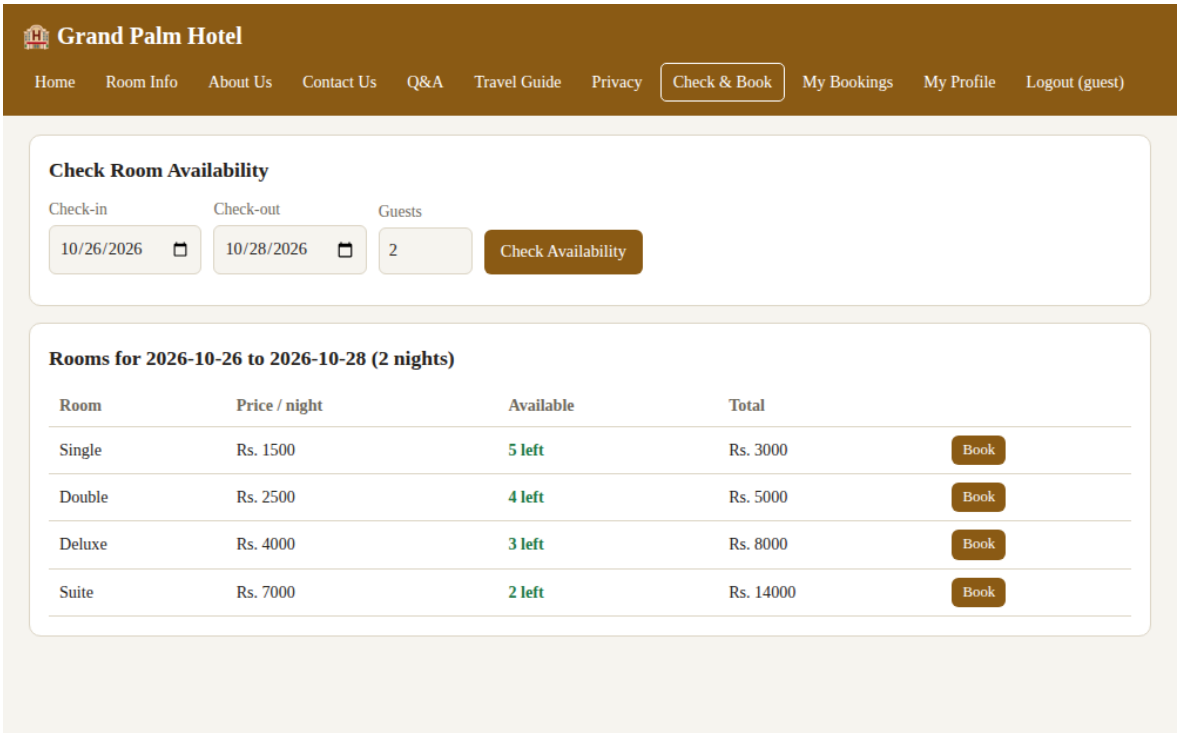


Fig 3: Guest - Check room availability for selected dates

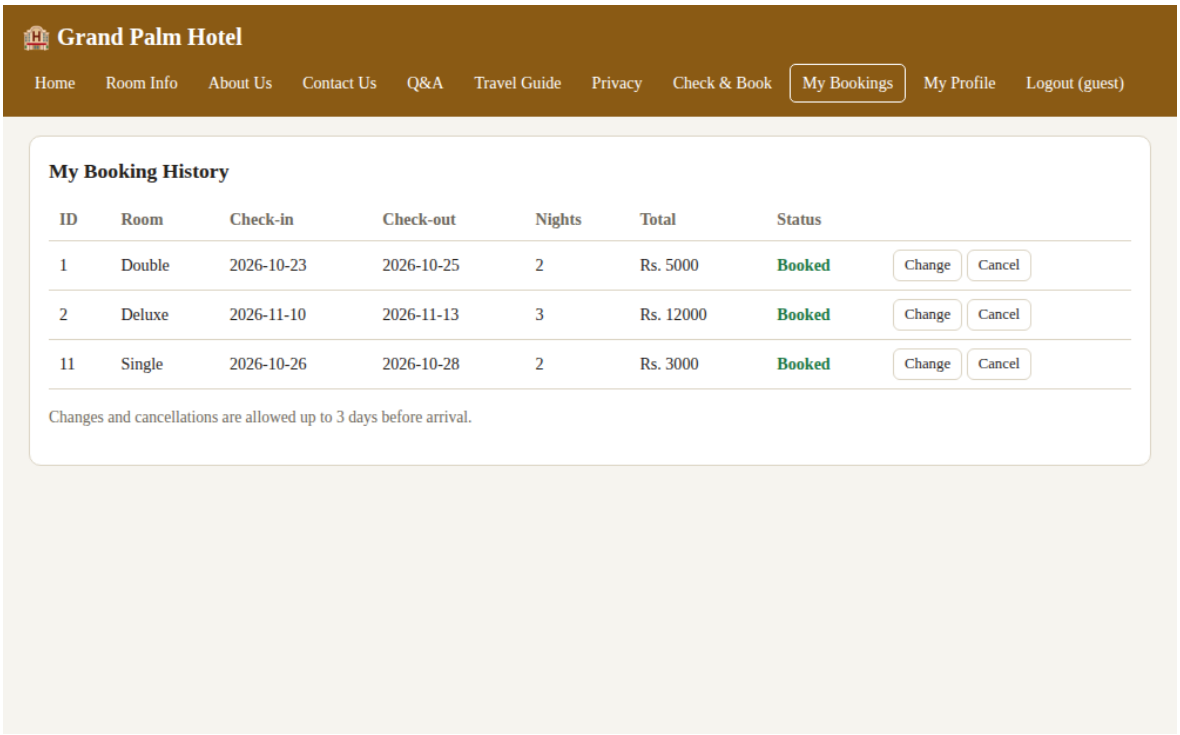


Fig 4: Guest - Booking history after booking a Single room

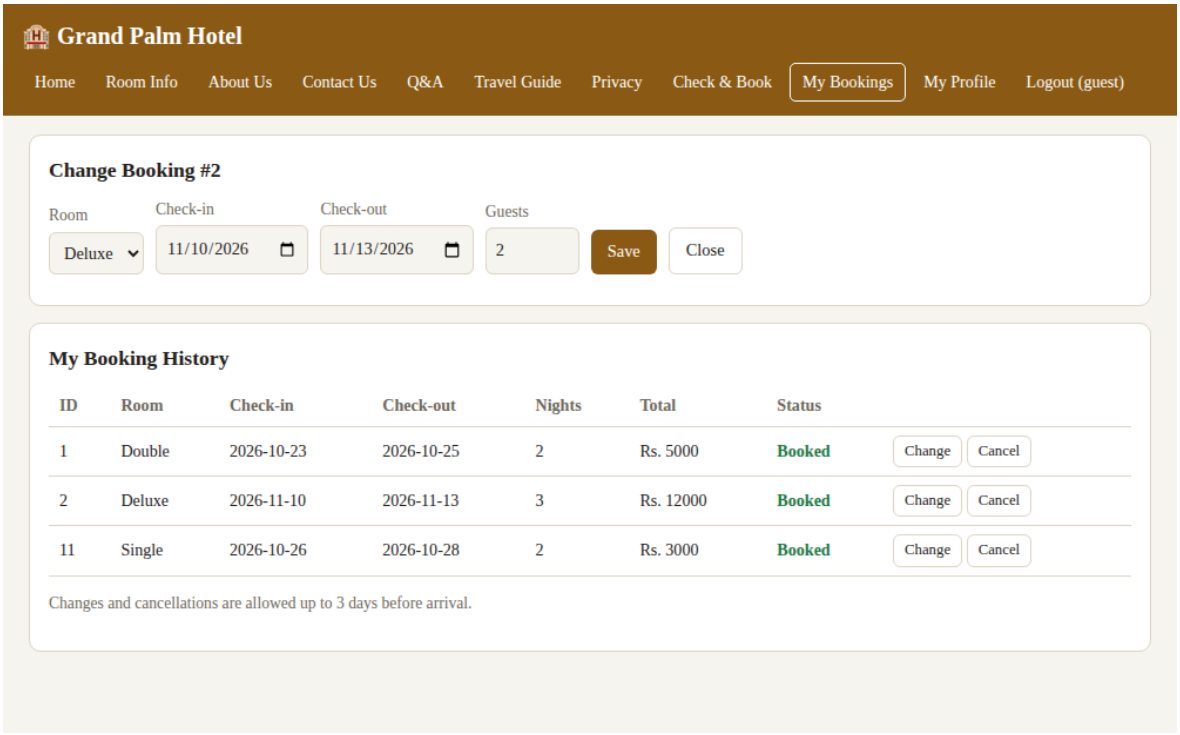


Fig 5: Guest - Changing a booking (cancelling booking #1, 2 days before arrival, is refused)

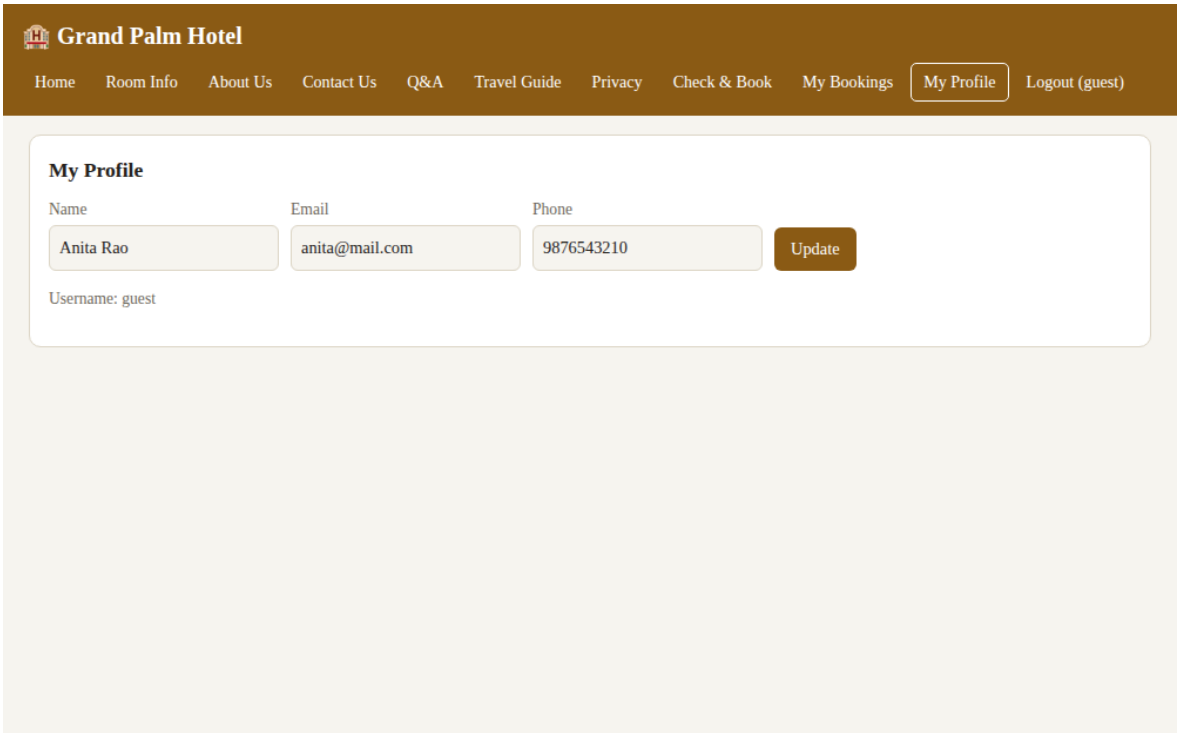


Fig 6: Guest - My Profile

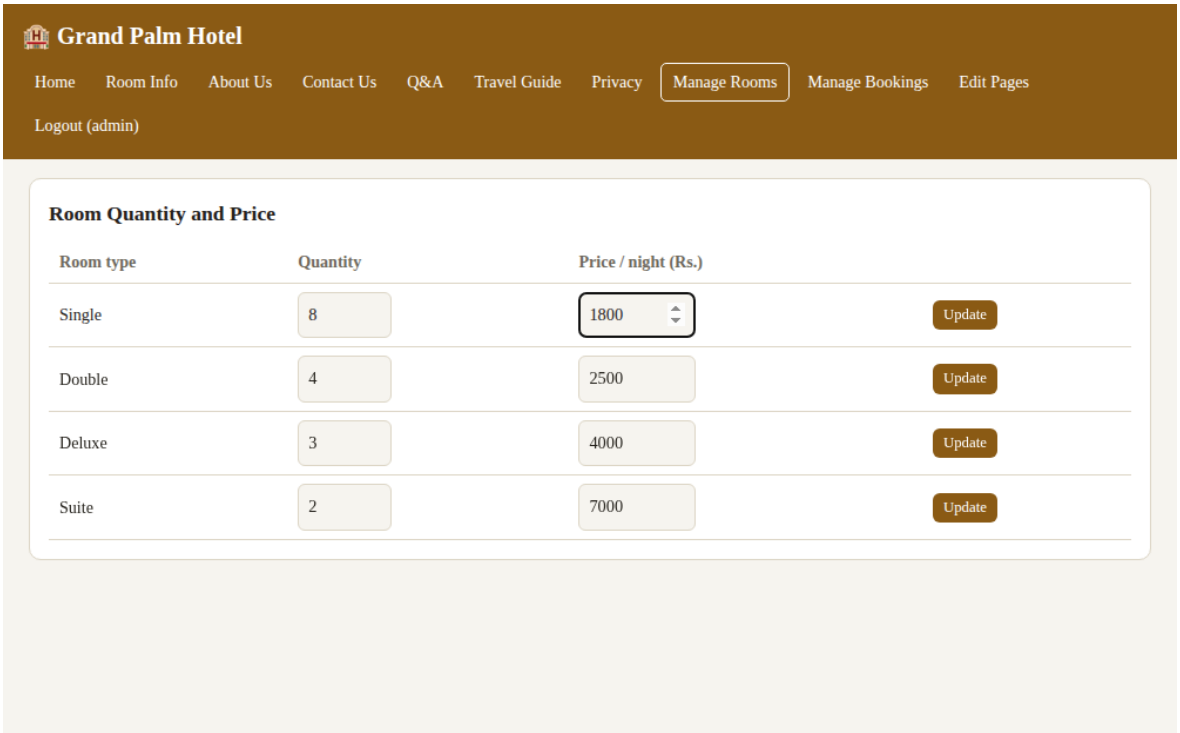


Fig 7: Admin - Update room quantity and price

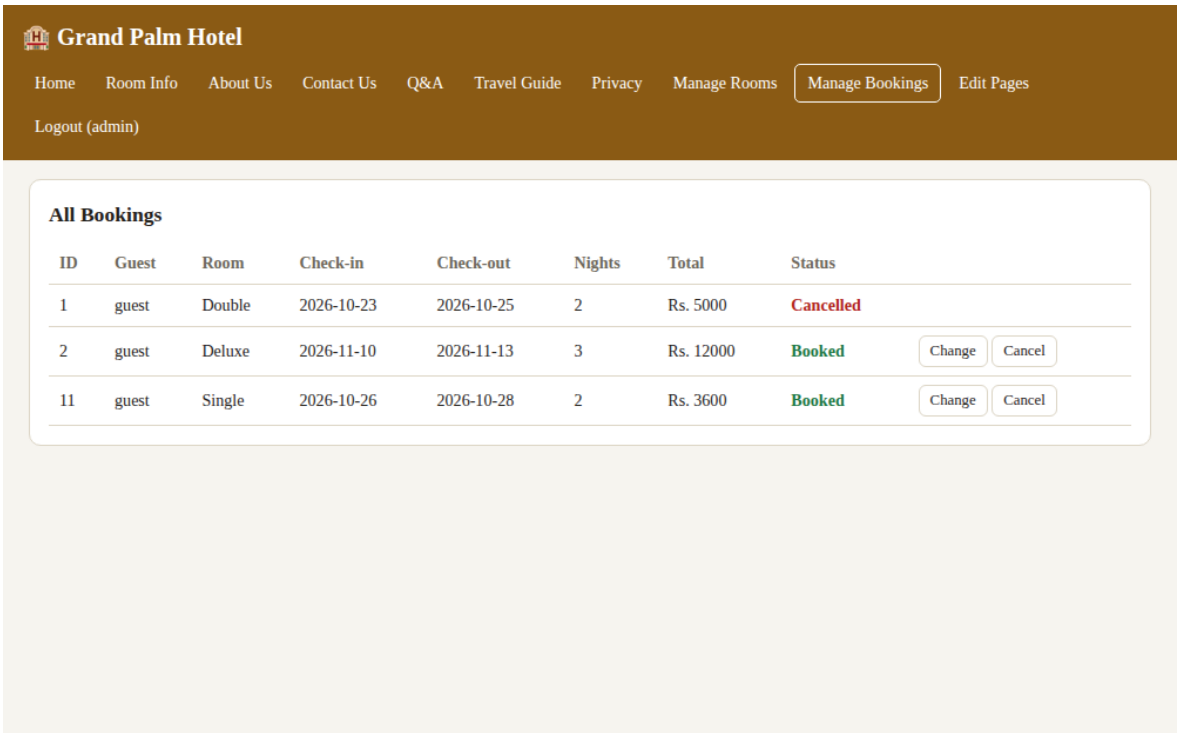


Fig 8: Admin - Manage bookings (booking #1 cancelled by admin)

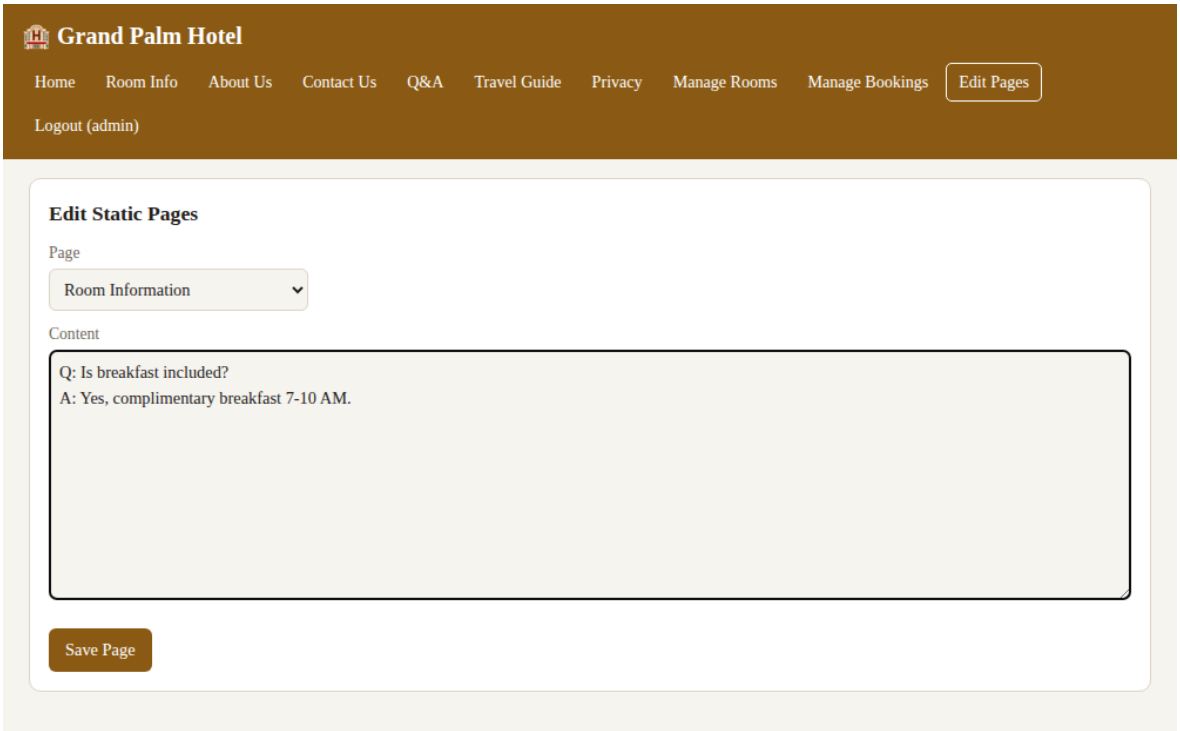


Fig 9: Admin - Edit static pages (Q&A updated)

Result

The online hotel room booking website was developed using HTML, CSS and JavaScript. Guest functions (register, profile, availability check, booking, history, change and cancel with the 3-day rule) and administrator functions (room quantity and price, booking change and cancel, static page editing) were executed successfully.